PRINCIPLES OF AGILE

A Guide to Iterative & Adaptable Development

Dr. Mohamed H. ElGazzar

Agenda

- 1. Introduction to Agile
- 2. The Agile Manifesto & Core Values
- 3. The 12 Principles of Agile
- 4. Popular Agile Methodologies (Scrum, Kanban)
- 5. Benefits and Challenges of Agile
- 6. Q&A

What is Agile?

More than a process, a way of thinking.

A Mindset & Methodology

Agile is a flexible, iterative approach to software development and project management, valuing individuals and interactions over processes and tools.

Core Focus Areas

- Iterative Development: Deliver value in small, frequent increments.
- Collaboration: Foster cross-functional teams and continuous feedback.
- Adaptability: Respond to change over following a rigid plan.

The Core Values (The Agile Manifesto)

Introduction to the foundational pillars of the Agile mindset.

We value:

- Individuals and interactions
- Working software
- Customer collaboration
- Responding to change

Over:

- processes and tools
- comprehensive documentation
- contract negotiation
- following a plan

People and Interactions Win

- Value individual team members and their interactions
- Team communication is key to effective problem-solving
- Human interaction addresses complex problems faster
- Prioritize people and interactions over rigid processes

Value 2 - Working Software over comprehensive documentation.

- Deliver functional software frequently
- Gather feedback from actual product usage
- Documentation should support, not replace, working features
- Primary measure of progress is working software

Value 3 - Customer Collaboration over contract negotiation.

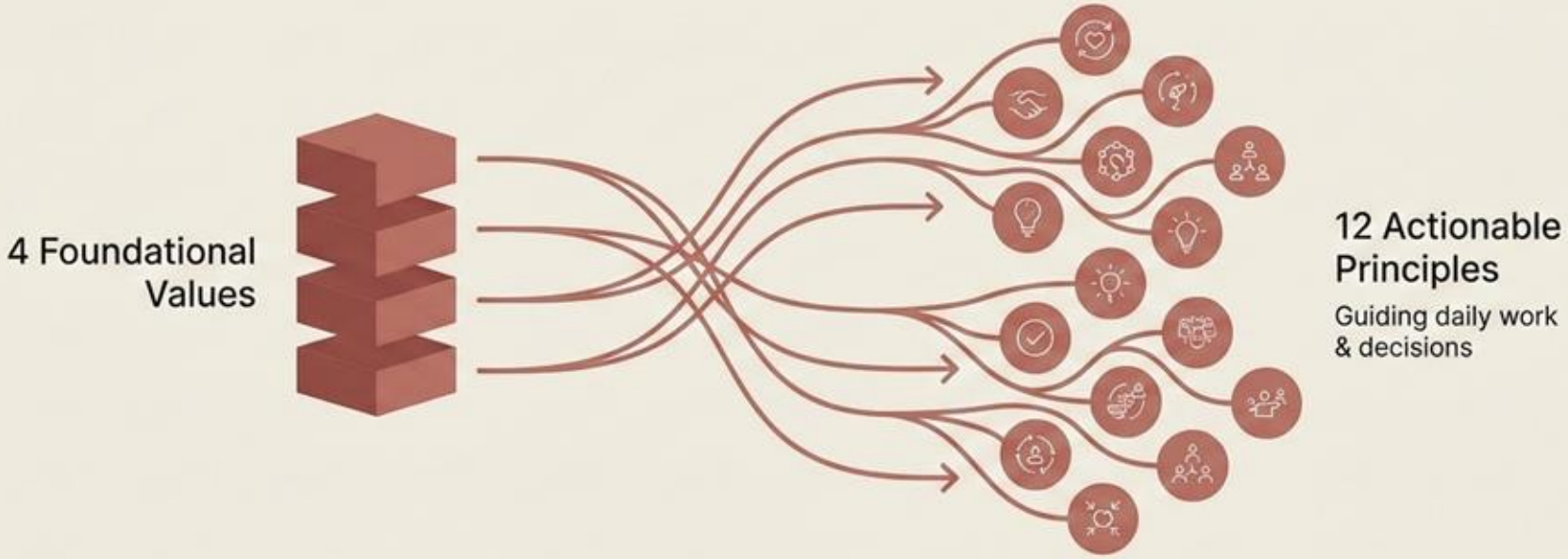
- Active user involvement is crucial
- Regular feedback loops build trust
- Shared understanding leads to better products
- Flexible contracts enable adaptation

Value 4 - Responding to Change over following a plan.

- Welcome changing requirements, even late in development
- Embrace change for customer's competitive advantage
- Plans should be flexible and adaptive
- Iterate and adjust based on feedback

Introduction to the Agile 12 Principles

How the 4 values translate into 12 actionable principles for day-to-day work



Dr. Mohamed H. ElGazzar

The Agile 12 Principles

Translating values into daily practices.

- Customer satisfaction through early, continuous delivery.
- Welcome changing requirements, even late.
- Deliver working software frequently.
- Business people and developers work together daily.
- Build projects around motivated individuals.
- Face-to-face conversation is best.
- Working software is the primary measure of progress.
- Promote sustainable development, constant pace.
- Continuous attention to technical excellence.
- Simplicity is essential.
- Self-organizing teams create best architectures.
- Regular team reflection and tuning.

Principle 1: Customer Satisfaction First

- Highest priority is customer satisfaction with early delivery
- Continuously deliver valuable products to the customer early
- Focus on early and continuous delivery for customer satisfaction

Principle 2: Welcome Change!

- Welcome changing requirements even late in development
- Agile processes harness change for the customer's competitive advantage
- Embrace changes to create a competitive advantage for customers
- Changes in requirements are welcomed, even late in the process

Principle 3: Deliver Frequently

01 — Deliver working software frequently from a few weeks to a couple of months

02 — The highest priority is to satisfy the customer through early and continuous delivery

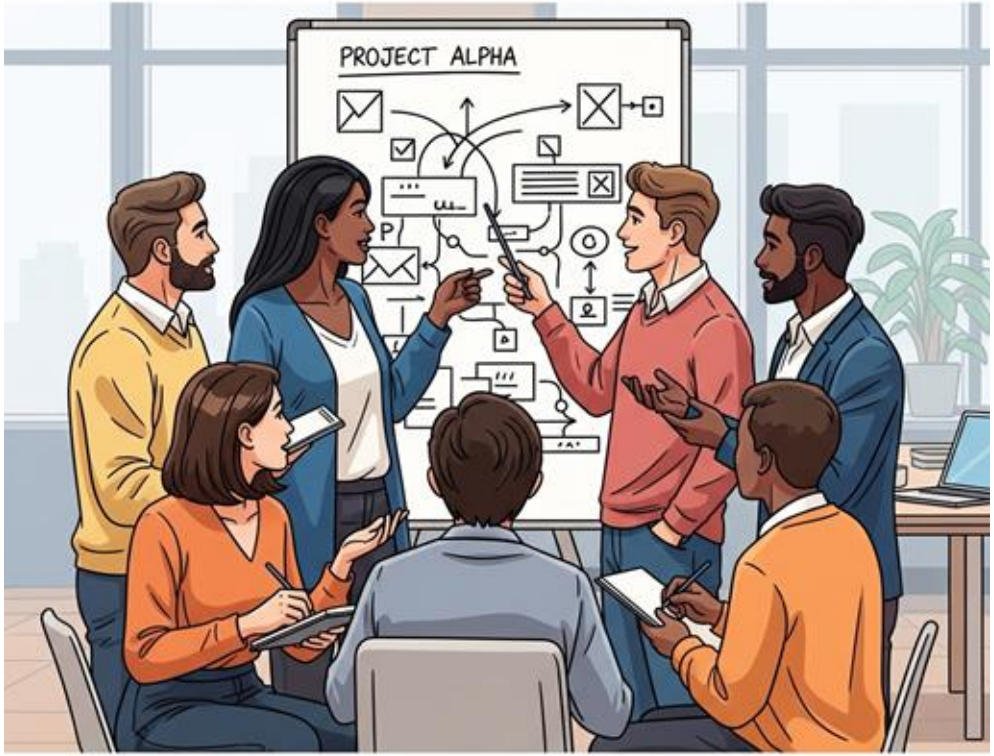
03 — Frequent delivery of software provides early and continuous value to customers

Principle 4: Work Together Daily

- Business and developers must work together daily
- Collaboration is essential throughout the entire project
- Daily interaction helps to maximize information sharing
- Consistent communication ensures better alignment and results

Principle 5: Motivate Individuals: Trust

- Build projects around motivated individuals for success
- Give people the necessary environment and support
- Trust the motivated individuals to get the job done
- The best architectures emerge from self-organizing teams



Principle 6: Face-to-Face Matters

- Conveying information face-to-face is the most efficient method
- Face-to-face conversation reduces ambiguities and misunderstandings
- Maximize the amount of important information shared quickly
- The most effective way to communicate within a development team

Principle 7: Working Product Progress

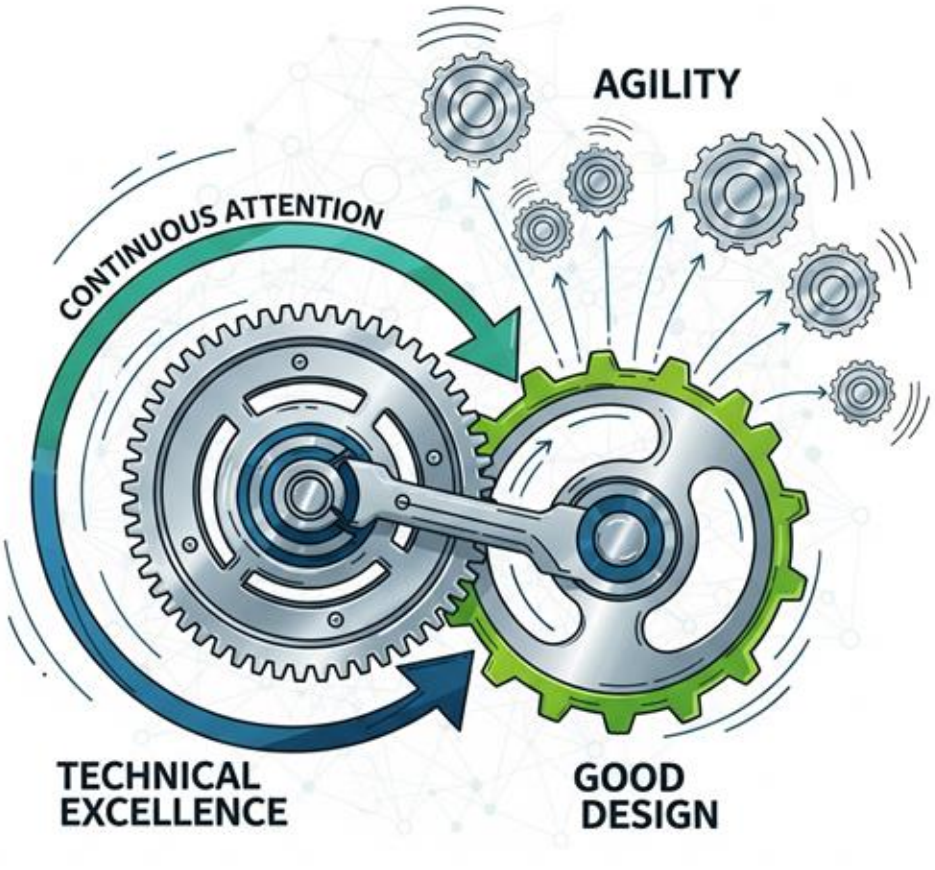
- Working software is the primary measure of progress.
- Focus on delivering a working product frequently and consistently.
- Delivery of value is superior to comprehensive documentation.
- Functioning solutions demonstrate actual progress toward goals.

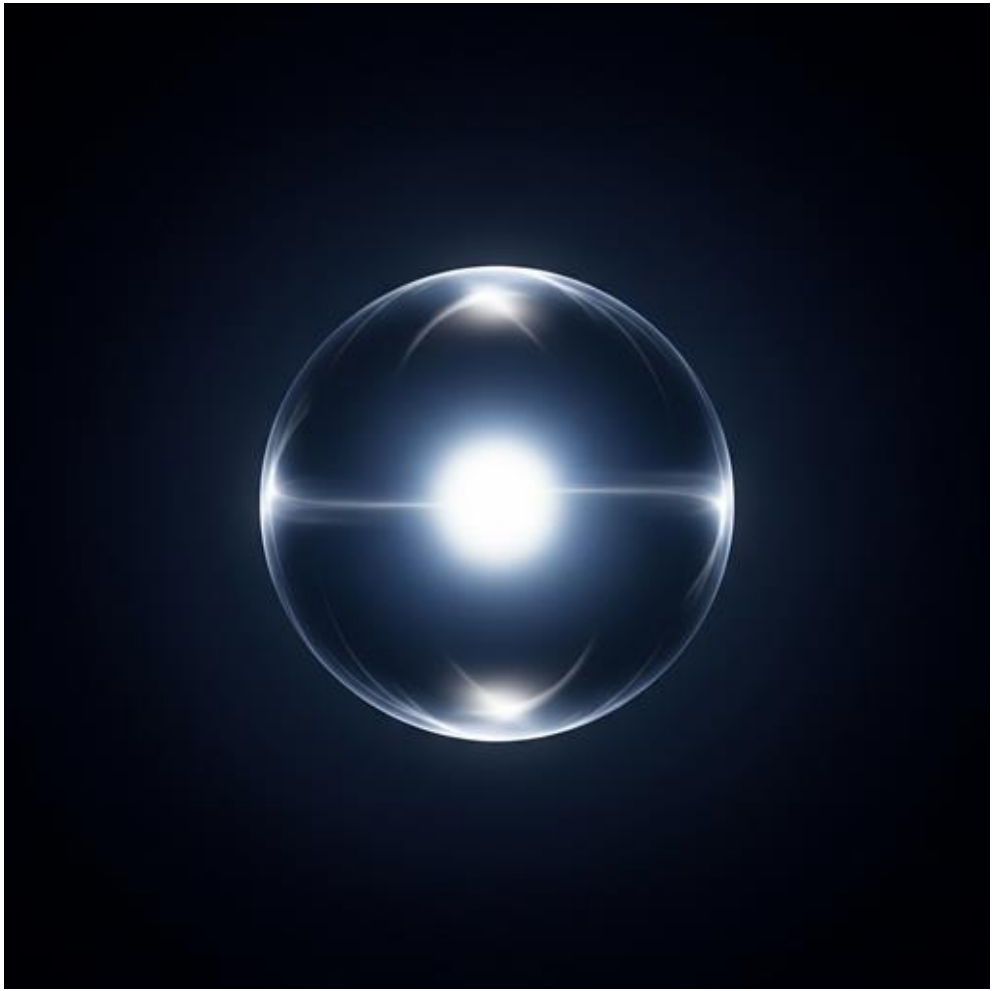
Principle 8: Sustainable Pace

- Development processes must maintain a constant and sustainable pace.
- Agile processes promote consistent and sustainable development indefinitely.
- Stakeholders and developers should be able to maintain the pace.
- A continuous pace ensures long-term and consistent delivery.

Principle 9: Technical Excellence Matters

- Continuous attention to technical excellence enhances agility
- Good design is crucial to maintaining technical excellence
- Focus on enhancing the design and technical quality consistently
- Sustained technical excellence simplifies future development efforts





Principle 10: Simplicity Is Essential

- Simplicity is essential for maximizing the amount of work not done
- Eliminate unnecessary complexity to enhance overall efficiency
- Focus on the essential elements that deliver the highest value

Principle 11: Self-Organizing Teams

- Best architectures, requirements, and designs emerge from self-organizing teams
- Empower teams to determine the best approach for reaching goals
- Trust in the motivated individuals to get the job done successfully
- Provide the necessary environment and support for team success

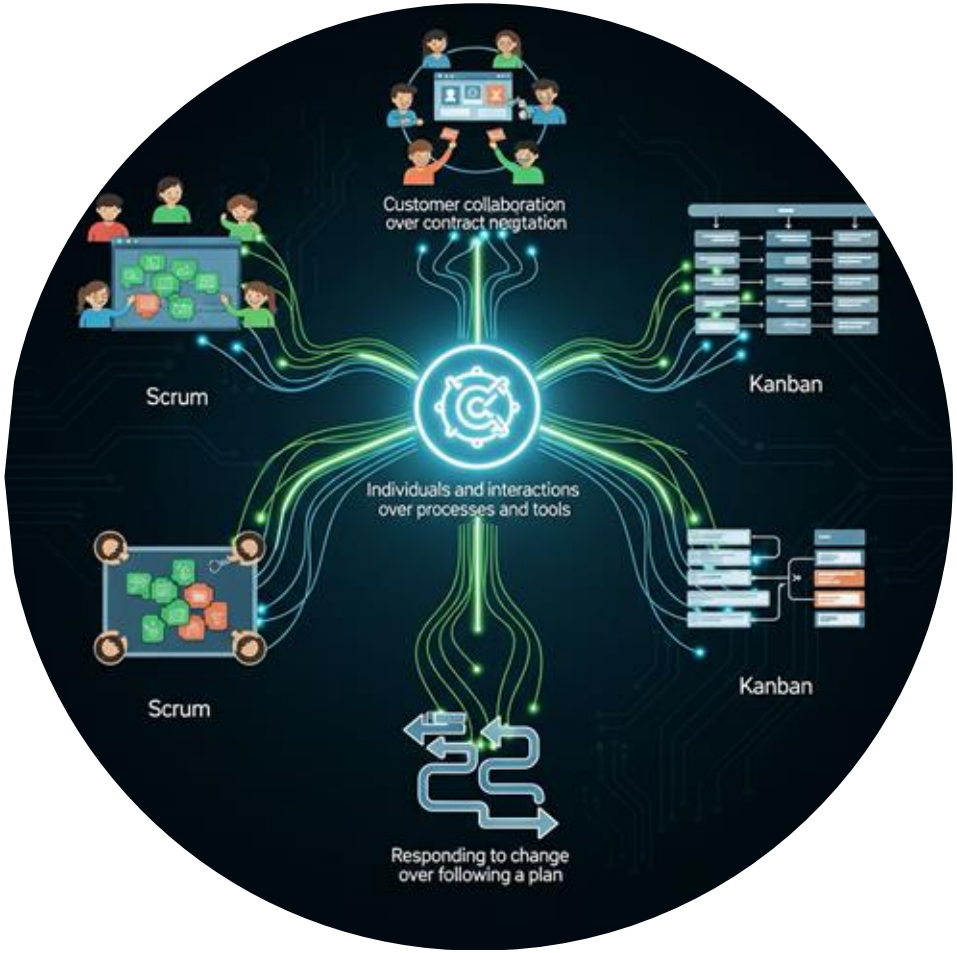
Principle 12: Reflect and Adjust

01 Regularly reflect on how to become more effective

02 Continuous improvement is vital for team success

03 Adjust behavior and processes based on team reflection

04 The team should fine-tune and tune its conduct



Agile vs Frameworks

- Agile principles guide the mindset and approach.
- Frameworks are specific methodologies like Scrum or Kanban.
- Principles are the core ideas of the Agile Manifesto.
- Frameworks offer concrete steps for implementation.

Scrum at a Glance

01 Scrum provides a framework for complex product delivery

02 Scrum consists of roles, events, and artifacts that guide the work

03 It delivers products using an iterative and incremental approach

04 The framework focuses on continuous improvement and adaptation

Kanban: What Is It?

- Kanban is a visual system to manage work in progress
- It focuses on continuous flow and limiting work in progress
- The system helps teams visualize their workflow and bottlenecks
- Kanban promotes continuous improvement and delivery of value

Scrum Versus Kanban

- Scrum is a specific iterative and incremental framework
- Kanban is a visual system for managing continuous workflow
- Scrum uses roles, events, and specific artifacts to guide work
- Kanban focuses on limiting work in progress and continuous flow
- Both frameworks promote continuous improvement and delivery of value

Extreme Programming Overview

- Extreme Programming is an Agile software development framework
- The key focus is delivering high-quality software quickly
- It emphasizes frequent releases and close customer collaboration
- Extreme Programming promotes simplicity and continuous feedback

Scrum Ceremonies in Action

- Daily stand-ups ensure alignment and share progress.
- Sprint planning focuses the team on the highest priorities.
- Sprint reviews showcase the delivered working software.
- Retrospectives allow the team to reflect and adjust processes.

Fake Agile Pitfalls

- Over-documenting everything instead of focusing on working software
- Excluding business people from daily development work
- Not delivering a working product frequently enough
- Failing to welcome changes late in development process

True Agile Mindset Benefits

- Get products to market faster for competitive advantage
- Improve team spirit and increase overall work engagement
- Minimize the risk of project failure and maximize success
- Increase customer happiness with continuous valuable delivery